

```
R3#
R3#enable
R3#
R3#configure terminal
Enter configuration commands, one per line. End with CTRL-Z
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#
R3(config)#exit
R3#configure terminal
*Aug 21 12:11:06.971: %SYS-5-CONFIG-I: Configured from console
R3#sh ip int brief
Interface IP-Address
Ethernet0/0 40.0.0.1
Ethernet0/1 30.0.0.2
Ethernet0/2 unassigned
Ethernet0/3 unassigned

R3#sh ip route
Codes: L - local, C - connected, S - static, D - EIGRP, EX - EIGRP external
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2, E1 - OSPF external type 1, E2 - OSPF external type 2,
i - IS-IS, su - IS-IS summary, ia - IS-IS inter area, * - candidate default, U - per interface default route, o - ODR, P - periodic download,
a - application route, + - replicated route, % - next hop unreachable
Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 20.0.0.0/8 [1/0] via 30.0.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.2/32 is directly connected, Ethernet0/1
40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 40.0.0.0/8 is directly connected, Ethernet0/0
L 40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

QEMU (Win-2)

```
C:\WINDOWS\system32\cmd.exe

Ping statistics for 40.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.0.0.01
Ping request could not find host ping. Please check the name and try again.

C:\>ping 10.0.0.10
Ping request could not find host ping. Please check the name and try again.

C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
    Reply from 10.0.0.10: bytes=32 time=4ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=3ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Ping statistics for 10.0.0.10:
        Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
        Approximate round trip times in milli-seconds:
            Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
    Ping statistics for 10.0.0.10:
        Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
        Approximate round trip times in milli-seconds:
            Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>
```

Device requires that you restart your system.

```
R3#
R3#enable
R3#
R3#configure terminal
Enter configuration commands, one per line. End with CTRL-Z.
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#
R3(config)#exit
R3#configure terminal
*Aug 21 12:11:06.971: %SYS-5-CONFIG_I
R3#sh ip int brief
Interface IP-Address
ooc1
Ethernet0/0 40.0.0.1
Ethernet0/1 30.0.0.2
Ethernet0/2 unassigned
Ethernet0/3 unassigned

R3#sh ip route
Codes: L - local, C - connected, S -
D - EIGRP, EX - EIGRP external
N1 - OSPF NSSA external type 1
E1 - OSPF external type 1, E2
i - IS-IS, su - IS-IS summary,
ia - IS-IS inter area, * - can
o - ODR, P - periodic download
a - application route
+ - replicated route, % - next

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 20.0.0.0/8 [1/0] via 30.0.0.1
30.0.0.0/8 is variably subnette
C 30.0.0.0/8 is directly connec
L 30.0.0.2/32 is directly connec
40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 40.0.0.0/8 is directly connected, Ethernet0/0
L 40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

QEMU (Win-1)

```
C:\WINDOWS\system32\cmd.exe
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=2ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
C:\>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=4ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms
C:\>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\>
```

Device requires that you restart your system.

```
R3#
R3#enable
R3#
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#
R3(config)#exit
R3#configure terminal
*Aug 21 12:11:06.971: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
-----
ooc1
Ethernet0/0              40.0.0.1        YES manual up          up
Ethernet0/1              30.0.0.2        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down

R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```


PNETLab | Topology

https://labs.surennetwork.com

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=2ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=4ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>_

device requires that you restart your
ect.

Win-1

10.0.0.10

e0

40.0.0.1

e0/0

Win-2

40.0.0.10

e0

Static Route

Physical

01 : 00 : 00

START

1

Search

freeSSHd update

C:\WINDOWS\system...

1:12 PM

ENG IN

17:42

21-08-2026

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ext
^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#
*Aug 21 12:08:36.656: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
-----
ooc1
Ethernet0/0              20.0.0.2        YES manual up          up
Ethernet0/1              30.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down

R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#
```



Search



ENG
IN



17:39
21-08-2026

```

R1
Ethernet0/2          unassigned      YES TFTP   down      down
Ethernet0/3          unassigned      YES TFTP   down      down

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#
*Aug 21 12:06:19.052: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface            IP-Address      OK? Method Status      Prot
-----
oob
Ethernet0/0          10.0.0.1        YES manual up          up
Ethernet0/1          20.0.0.1        YES manual up          up
Ethernet0/2          unassigned      YES TFTP   down      down
Ethernet0/3          unassigned      YES TFTP   down      down

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       40.0.0.0/8 [1/0] via 20.0.0.2
R1#

```


 R1

▶ Win-1

Physical

01 : 00 : 00 START